After the Decision: Learning to Adapt from Delayed Outcomes

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Abstract

Scientific agents often choose among several tools that return plausible but conflicting answers. The challenge is not only selecting a provider before its output is visible: correctness feedback is produced after termination, may be delayed, and must update later choices without inventing labels for unselected tools. We present MIRA-Relay, an outcome-aware routing loop that combines a frozen discriminative provider surface with a lightweight provider-local residual adapter and an explicit release ledger. At cold start the adapter is exactly the frozen selector; after a selected result is released, it updates only that provider's residual reliability and can move the next action. We evaluate a one-call BBB routing contract with public truth, four checkpoint-bound fresh providers, scaffold-disjoint confirmation molecules, selected-only feedback, and delays $D \in \{0, 1, 4\}$. On the natural fresh holdout, MIRA matches Frozen at 89.81

CCS Concepts

• **Computing methodologies** → **Multi-agent planning**; *Reinforcement learning*; Machine learning.

Keywords

tool selection, online learning, delayed feedback, scientific agents, molecular property prediction

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1 Introduction

A scientific agent must decide which tool's prediction to use when several providers estimate the same property. That choice matters even when every call completes: execution success establishes that an answer was returned, while usefulness depends on whether the answer is correct for the input. Molecular property prediction makes this distinction measurable because provider outputs can be checked against independently recorded endpoint labels.

Valid responses can disagree on the same molecule. In our frozen BBB development panel, molecule `z000282` has benchmark label 1: `admet_ai_1_4_0` returns 0, while `b3clf_dtree_common` returns 1, both with successful runtime status. The example establishes a selection problem; it does not establish which provider is best on the next molecule. During evaluation, the agent sees neither unselected outputs nor the current correctness label before choosing.

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Outcome feedback creates two obstacles: sparse coverage and delayed release. A fixed policy cannot use new evidence. A selected-only learner sees feedback only from chosen providers, leaving alternatives sparsely observed; delay postpones when feedback is usable. These constraints motivate our transfer hypothesis but do not prove that any bandit fails. We therefore compare matched online controls and test whether sharing changes later decisions.

We hypothesize that a released outcome can inform later choices through a shared released state. Each provider keeps its own residual, so the update need not assume that providers are independent or interchangeable. An immutable fit-time reference anchors the comparison. The claim requires a change in later exploitation under identical observations.

MIRA-Relay is the concrete instantiation of this hypothesis within the MIRA adaptation framework. Its cold-start surface is frozen, while a selected-only reliability ledger adds a provider-specific residual only after the corresponding outcome is released. The residual is centered on that provider's fit-time success rate, so the adapter starts exactly at the frozen policy and changes only when released evidence is strong enough to move a later ranking. Runtime failure, unknown truth, and chemical error remain separate ledger fields. A post-call verifier is a defined extension interface; the paper's primary claim is the one-call routing contract.

This draft makes four testable commitments, each paired with a falsification condition in the experimental matrix:

- **Task validity.** Public truth and a concrete provider panel yield input-dependent headroom over a fixed provider without leaking unselected outcomes; this fails if an evaluator-only panel score is the only source of signal.
- **Faithful adaptation.** Selected-only delayed events can be replayed with exact visibility, release clocks, and failure semantics; this fails if the contract/replay verifier cannot validate the recorded action, release, and serialized-state chain.
- **Mechanistic adaptation.** Released context-relative feedback changes later exploitation decisions beyond Frozen and matched online controls; this fails if the changes disappear under the declared drift or cannot be traced to released selected-only events.
- **Practical value.** Any improvement in future correctness must survive the strongest legal fixed and matched online controls while reporting harmed decisions, calls, latency, and memory; otherwise the conclusion is limited to a protocol or diagnostic result.

The remainder of the paper defines the task, gives the proposed specification, fixes the benchmark and baseline contracts, and reports the natural fresh-menu confirmation together with a preregistered correctness-drift stress.

2 Problem and Task Contract

We study one agent making a sequence of tool calls. Figures 1 and 2 show the same contract at two scales: the first separates the learned relay from execution and evaluation, while the second fixes the

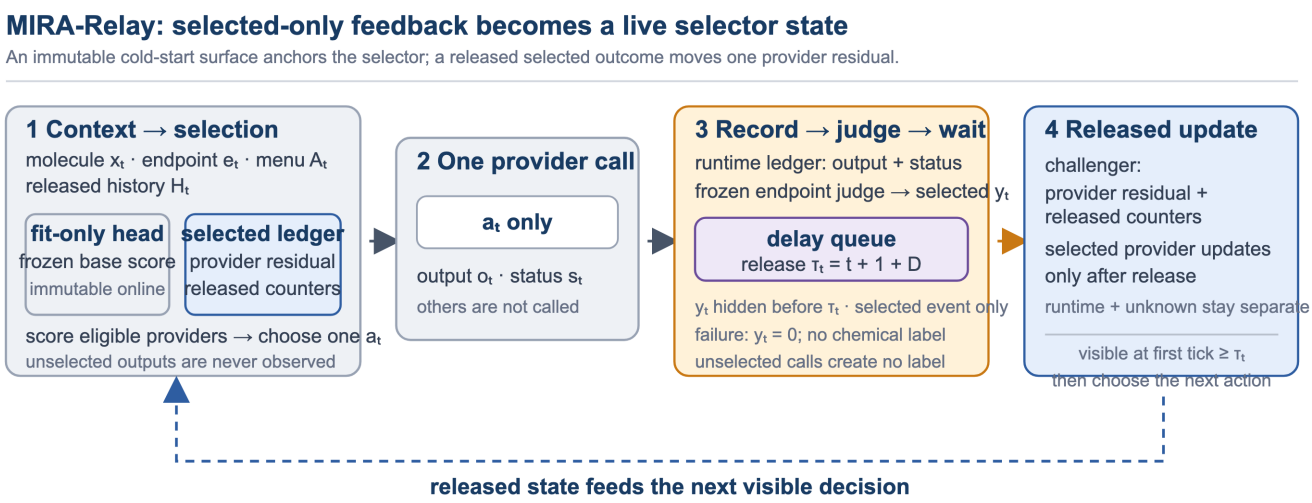


Figure 1: MIRA-Relay closes the loop from a pre-call choice to a later decision. Molecule, endpoint, eligible menu, and released state enter a frozen provider surface plus a provider-local residual ledger. One selected call produces a terminal record; the frozen endpoint judge supplies chemical correctness only after termination, while runtime failure follows its own ledger. The selected outcome waits until $\tau_t = t + 1 + D$ and then updates the released state; decisions continue while feedback is pending.

event order that determines when feedback can affect a later action. The contract is defined at the granularity of a concrete provider, not a provider family or a post-hoc majority vote.

2.1 Episode, menu, and feedback

At decision tick t, the agent receives a molecule x_t, an endpoint e_t, and an available menu A_t of concrete provider identifiers. It chooses one provider $a_t \in A_t$ before reading the current output panel. The benchmark stores an independent endpoint truth $y_t \in \{0, 1\}$ for evaluation. The selected call returns an output o_t, a runtime status $s_t \in \{\text{ok}, \text{error}\}$, and, when available, a confidence value c_t. We keep the labels that drive the method separate:

$$r_t^{\text{run}} = \mathbf{1}[s_t = \text{ok}],$$
$$r_t^{\text{chem}} = \begin{cases} \mathbf{1}[o_t = y_t], & s_t = \text{ok}, \\ \text{null}, & s_t = \text{error}, \end{cases}$$
$$r_t^{\text{sys}} = \mathbf{1}[s_t = \text{ok} \wedge o_t = y_t].$$

The correctness head is trained only on non-null r^{chem}; the runtime head uses r^{run}; a bandit reward may use r^{sys} with separate failure and chemical-error costs. Thus an execution failure is not silently relabeled as a chemical mistake.

After termination, a replay judge records the selected outcome. The outcome becomes visible only at its registered release tick τ_t. The legal history at tick t is

$$H_t = \{(x_s, e_s, a_s, r_s^{\text{sys}}, \tau_s) : s < t,\ \tau_s \le t\}.$$

For example, for molecule `z000282`, the benchmark truth is 1. The call to `admet_ai_1_4_0` returns 0 with status ‘ok‘. The call to `b3clf_dtree_common` returns 1 with status ‘ok‘. Both calls have $r^{\text{run}} = 1$, but their chemical labels differ. A decision at tick 2 cannot read either unselected output, the current truth, or any future terminal record.

Provider parameters use a registry $\mathcal{P}$ fixed at initialization; $A_t \subseteq \mathcal{P}$ is the current candidate menu. Removing a challenger leaves its stored state intact. For the historical BBB replay, $|\mathcal{P}| = 33$ throughout the stream. The queue orders releases by due tick and breaks ties by enqueue order; each feedback is consumed once and receipts record the release order explicitly.

For the fixed-delay replay, a decision at tick t terminates, enqueues its feedback, and releases it at

$$\tau_t = t + 1 + D, \qquad D \in \{0, 1, 4\}.$$

Only after this release may the event enter $H_{t'}$ and affect a later decision. For $D = 0$, the release is processed before the next decision tick and never changes the action that caused the event. Unknown truth creates no correctness label and stops the current scoring stream; a refusal creates no selected-provider label. Both remain explicit in the episode ledger.

2.2 Selection and verification interfaces

The pre-call selector and post-call verifier have disjoint interfaces:

$$a_t = \pi_{\text{sel}}(x_t, e_t, A_t, H_t, S_t), \qquad v_t = \pi_{\text{ver}}(o_t, c_t, s_t),$$

where v_t is one of accept, escalate, or refuse. The verifier can request a second provider only after the first call returns. This is an extension interface for E5, not an action performed by the current direct-selection runner. In the running example, a selector may choose ‘b3clf_dtree_common‘; only after its output and confidence are available can the verifier decide whether to escalate.

The online learner maintains a serializable post-release state S_t. At tick t, the release phase first applies exactly the events due at that tick; the selector then reads the resulting state:

$$S_t = U\left(S_{t-1}, \{(x_s, e_s, a_s, r_s^{\text{sys}}, \tau_s) : s < t,\ \tau_s = t\}\right),$$
$$a_t = \pi_{\text{sel}}(x_t, e_t, A_t, H_t, S_t).$$

With no due event, U is the identity. Thus a $D = 0$ event from tick t enters S_{t+1} and can affect action $t + 1$, but never action t. No correctness-residual update is allowed for an unselected provider, an unreleased event, an unknown label, or a refusal. A terminal unknown or refusal may still update runtime/availability bookkeeping; every selected scored provider, including the fit-best provider, receives only its own released residual update. This contract lets a receipt verifier check the visible read set, action, release queue, recorded state fields, and schema/counts. Complete metric recomputation is left to the E5 transition replayer; this receipt check does not by itself recompute scores, propensities, or the update.

2.3 Estimands and costs

The primary estimand is prequential system success: a selected call must return a valid and correct answer. We also report chemical correctness conditional on valid returns, paired rescued and harmed decisions, failures, and call and update costs. The extension interface additionally accounts for escalation, refusal, and unknown truth; those utilities are outside the reported one-call result. Static full-panel accuracy and an evaluator-only retrospective best-fixed control measure task headroom separately from online effects.

3 MIRA-Relay: Proposed Method Specification

This section specifies MIRA-Relay and the information boundary tested in the one-call experiments.

3.1 Cold-start scores and the immutable reference

Let $\phi(x) \in \mathbb{R}^d$ be a frozen molecule representation and let $K = |\mathcal{P}|$ be the number of concrete providers. For each provider a, a logistic correctness head w_a^0 is fitted once on the fit partition. Its cold-start score is

$$q_a^0(x) = \sigma\left((w_a^0)^\top \phi(x)\right).$$

The current menu only masks which coordinates can act; no current output is read before the action. For the fresh BBB confirmation, $d = 65$: 64 Morgan bits plus an intercept. The fit-only Best Fixed provider is the provider with the highest fit-partition correctness. The frozen base scores are immutable throughout evaluation, and every provider switch is still counted even when it is wrong.

The running case is concrete. For molecule `z000282`, $\phi(x)$ is the frozen 65-dimensional vector built from its canonical SMILES and the intercept; the fit heads assign separate scores to the four fresh providers. At cold start, MIRA-Relay and Frozen therefore choose the same provider for the same menu. Providers outside the sealed checkpoint manifest are out of scope.

3.2 Context-relative reliability adapter

For provider a, let $\bar{r}_a$ be its fit-partition correctness rate and let $n_{a,t}$ and $s_{a,t}$ be, respectively, the number of its selected outcomes released by tick t and the number that were correct. With a fixed prior count $n_0 = 20$, the released posterior rate and residual are

$$\widehat{r}_{a,t} = \frac{n_0 \bar{r}_a + s_{a,t}}{n_0 + n_{a,t}}, \qquad \Delta_{a,t} = \widehat{r}_{a,t} - \bar{r}_a.$$

The action score is

$$q_{a,t}(x) = q_a^0(x) + \Delta_{a,t}, \qquad a_t = \arg\max_{a \in A_t} q_{a,t}(x_t).$$

At cold start, $n_{a,0} = s_{a,0} = 0$ and therefore $\Delta_{a,0} = 0$: the adapter is exactly Frozen. When a selected outcome becomes visible, only that provider's counters change; an unselected provider never receives a synthetic label. A runtime error is recorded as an execution failure and is not inserted into the chemical-correctness count. This provider-local residual is added to the x-conditioned frozen score. Here "context-relative" refers to the current provider ranking and released evidence state, not to a second x-conditioned head; the fit-time representation itself never changes.

The case makes the update concrete. If the selected provider for `z000282` is correct, $s_{a,t}$ and $n_{a,t}$ each increase by one at its release tick. If it is wrong, only $n_{a,t}$ increases. The next score changes by the corresponding residual; all unselected providers keep their previous residuals. The update is $O(1)$ per released event and scoring is $O(Kd)$, while the online ledger stores $2K$ counters and the fixed heads remain outside the online state.

3.3 Bounded evidence interface

The same selector can be extended after the first call. Let E_t denote the returned output and runtime record. A verifier may accept, refuse, or request one additional provider call by minimizing a declared loss that includes call cost and refusal cost. This interface is specified so that a future evidence experiment has a fixed action vocabulary; it is outside the primary one-call result reported here.

3.4 Release ledger and delayed update

The online state is the selected-only ledger $S_t = \{(n_{a,t}, s_{a,t}) : a \in \mathcal{P}\}$. At tick t, the release queue first applies every event with due time $\tau = t$; the selector then reads the resulting state. For a fixed delay,

$$\tau_t = t + 1 + D, \qquad D \in \{0, 1, 4\}.$$

Thus a $D = 0$ event from tick t can affect action $t + 1$, never action t. The state transition is deterministic given the released event prefix, which makes the receipt independently replayable.

The update touches only the selected provider. For a scored terminal record with chemical label $r_t^{\text{chem}} \in \{0, 1\}$,

$$n_{a_t,t+1} = n_{a_t,t} + 1, \qquad s_{a_t,t+1} = s_{a_t,t} + r_t^{\text{chem}},$$

with all other provider counters unchanged. If the call fails, r^{chem} is null and the runtime-failure ledger records the event separately. Unknown truth and refusal do not update correctness counters. This separation is the central information boundary: the learner receives exactly the selected outcome after release, never the current truth or an unselected provider's output.

The update is $O(1)$ per released event and scoring all K providers costs $O(Kd)$. The online state uses $2K$ counters plus masks and the release queue. For `z000282`, a released result changes one provider residual, so the next action can switch according to the evidence.

3.5 Action rule and falsification

The primary one-call runner uses the highest current score in the eligible menu. There is no result-dependent probe in the reported fresh confirmation, so every method receives the same one-call budget and the same selected-only feedback clock. The controlled stress flips the selected label of the fit-best provider after the preregistered midpoint; it does not reveal any unselected output or alter the input features.

The implementation preserves five invariants: (i) no current truth or unselected output is read before action, (ii) only released selected events update state, (iii) the fit surface is immutable, (iv) runtime failure and chemical correctness remain separate, and (v) a serialized state snapshot plus event prefix deterministically reconstructs the next action. A contract/replay verifier checks action, release queue, serialized state fields, and event schema, then replays every action and score using the recorded runner primitives.

The adaptive claim is rejected if the residual cannot change a later action under the declared drift, if contract/replay checks find a mismatch, or if the gain disappears against the same-schedule Frozen and matched Online-SGD controls. Action changes alone are not treated as utility; the evaluation reports rescued and harmed decisions, scaffold-cluster intervals, calls, and update latency.

4 Benchmark and Baselines

4.1 Two-layer benchmark contract

The primary benchmark has a public truth layer and a provider routing layer. The truth layer uses the Therapeutics Data Commons (TDC) BBB endpoint and its fixed label semantics and split archive [5]. The development manifest contains 722 BBB molecules with reference labels; 716 have complete cached outputs from the 33-provider Task36 panel. Canonical conflict groups are excluded under the project decision while raw conflict receipts remain available for audit. The old development panel overlaps the official BBB test by 147 canonical molecules and 235 scaffolds, so it cannot be called an untouched confirmation set.

The routing layer turns cached provider outputs into sequential decisions with a fixed registry, a dynamic eligible-menu mask, selected-only feedback, delayed release, and explicit failures. It is a project protocol grounded in public truth. Our primary study uses the fixed provider registry; 716 of 722 development molecules have 33 runtime-ok cells, while six molecules retain 144 explicit error cells. Eligible challengers may change through runtime gates. Provider arrival and version migration require separate protocols. Unknown truth stops the current single-call stream; verification and refusal require the separately specified E5 protocol. hERG and AMES remain extension targets pending endpoint and lineage audits.

4.2 Baselines and fairness

Table 1 fixes the role of each comparator. All deployable methods receive the same molecule, endpoint, menu, feature stratum, initialization permission, selected-only feedback, delay, and call budget. Full-panel labels are allowed only in fit-only warm starts, the evaluator-only retrospective best-fixed control, or offline audits. That control is never included in a deployable ranking.

Table 1: Baseline roles under the common routing contract. A row marked "adapted" is reported separately from strict matched comparisons until its likelihood and prior semantics are identical.

Role	Comparator	Question answered
Lower anchor	Random	Is there signal beyond blind selection?
Fixed anchor	Best Fixed	What does a fit-only incumbent achieve?
Retrospective fixed control	Best Fixed on confirmation	How much headroom remains for a deployable selector?
Static control	Frozen selector	Does representation plus fit-time mapping suffice?
Matched online	Online SGD	Does ordinary selected-only updating help under the strict contract?
Separate control	Replay SGD (separate-control)	Does a prior-compatible historical replay provide a useful secondary check?
Contextual control	Greedy LinUCB	Does standard optimism help under the same contract?
Modern adapted	BayesianRouter; PILOT audit-only	How does a recent router transfer when adapted faithfully?
Evidence controls	Independent / Frozen Joint (out of scope)	Does provider dependence or online updating add value beyond the same evidence model?
Audit-only modern	GraphRouter (audit-only; currently blocked)	Can graph routing be audited without post-hoc tuning?
Task controls	Direct property predictor; confidence threshold (out of scope)	Is gain from routing or from recovering the endpoint / post-call risk?
Proposed method	MIRA-Relay (MIRA-Context)	Does released provider-local evidence improve later selection under declared drift?

The fairness record must include source commit, checkpoint or sealed-fit-state hash, feature and prior, fit-only warm start, feedback likelihood, delay, seed, budget, selection/update latency, state bytes, failure and unknown denominators, and license or audit-only status. Missing fields remain missing; an adapted baseline is not promoted by relabeling it as native.

5 Experimental Protocol and Matrix

The experiments are organized by questions rather than by implementation order. Every result uses raw JSONL, a configuration and input hash, per-event visibility and release fields, a source-tree

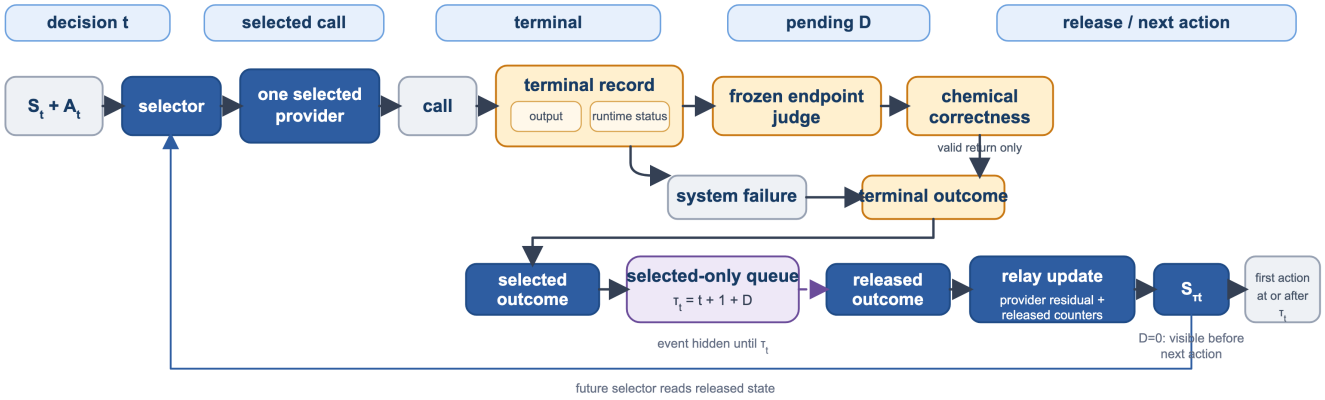


Figure 2: The terminal feedback contract. At decision tick t, the selector sees S_t and A_t and chooses one provider before any current output is observed. The terminal record is split into output and runtime status; the frozen endpoint judge then yields chemical correctness only after termination. Runtime failure and chemical correctness form the selected terminal outcome, which waits in a selected-only queue until $\tau_t = t + 1 + D$. The relay update is visible to the first action at or after release; for $D = 0$, this is the next action.

identity, and a contract/replay verification script. Results are aggregated by molecule and scaffold cluster; individual event rows are not treated as independent samples.

5.1 Research questions

RQ1 (validity): Does the frozen panel contain context-dependent headroom over the fit-only incumbent?

RQ2 (comparison): Under selected-only delay, does MIRA-Relay improve future correctness at matched cost?

RQ3 (mechanism): Does a released provider residual change later exploitation under the declared drift?

RQ4 (boundary/cost): How do delay, failures, exposure, verification, latency, memory, and confirmation affect the conclusion?

MIRA denotes the adaptation framework and MIRA-Relay is the implementation specified here, with MIRA-Context as its dynamic residual adapter and MIRA-Relative retained as a stopped negative variant. Strict common-prior controls share its evaluation contract; historical Replay and adapted routers retain separate comparison strata.

5.2 Pre-registered matrix

Table 2 records the preregistered matrix. The fresh one-call confirmation and controlled drift stress are now frozen and reported below; post-call verification and the wider 33-provider model-fresh extension remain outside this paper claim.

E2a is the strict common-prior comparison; E2b retains Replay as a separate historical control, and E2c reports adapted likelihood/prior strata separately. E3 and E4a are two views of the same frozen controlled-stress receipt: delay may still change actions, so contrasts are whole-policy paired effects. Family-macro results account for related providers. E4b is the bounded evidence extension and must use the same call budget across methods. Natural arrival, retirement, version change, and drift remain BLOCKED without a frozen trace.

6 Results

The reported result views use the closed fresh-menu receipts. Extension cells remain explicitly outside the declared one-call claim.

Confirmed fresh-menu result. The primary confirmation contains 270 natural streams and 120 controlled-drift streams. The natural stream has no runtime failures, unknowns, refusals, or extra calls; the controlled stress changes only the evaluator-side selected label after tick 78. Raw events and contract/replay verification are listed in the artifact paths in the supplementary material.

Natural confirmation and controlled drift. On the 157-molecule fresh holdout, Frozen, MIRA-Relative, and MIRA-Context all obtain 89.81

7 Error and Cost Analysis

We report errors by provider, scaffold group, runtime status, and feedback visibility. A wrong terminal prediction, a provider execution failure, an unknown truth, and an intentional refusal are different events and remain in separate denominators. The current fresh receipt reports selected correctness, rescued/harmed switches, calls, and update timing; the post-call utility decomposition is reserved for the out-of-scope evidence extension. This separation is necessary because a small pooled accuracy change can hide a large number of harmful switches.

The cost table includes calls, selection/update p50 and p95, serialized state, and replay time. Replay time measures policy computation; it is not provider money, tokens, or service latency. Post-call verification is outside scope.

Table 2: Pre-registered experiment matrix and closure status. VERIFIED cells have raw event receipts and contract/replay verification; OUT OF SCOPE cells are not used in the paper claim.

ID / RQ	Setting	Comparators / intervention	Primary measurements	Status
E1 / RQ1	BBB development, scaffold folds, fit-only panel	Random, Best Fixed, retrospective best-fixed, Frozen	OOF correctness, headroom, family macro, conflict and failure counts	VERIFIED
E2a / RQ2	fresh four-provider menu, delay $\{0, 1, 4\}$	Frozen, Matched Online SGD, MIRA-Relative, MIRA-Context, Greedy, BayesianRouter	prequential correctness, paired CI, calls, update p50/p95	VERIFIED
E2b / RQ2	historical 33-provider receipt	Replay SGD only against its own historical Frozen/control	separate prior-lineage diagnostic	OUT OF SCOPE
E2c / RQ2	fresh menu, adapted online likelihoods	LinUCB and BayesianRouter	within-stratum paired effect and cost	VERIFIED
E3 / RQ3	same controlled drift receipt; mechanism view	Frozen, Matched SGD, Greedy LinUCB, MIRA-Context	action changes, rescued/harmed, scaffold CI	VERIFIED
E4a / RQ4	same controlled drift receipt; outcome view	fit-best selected labels flip at midpoint; Frozen, Matched SGD, LinUCB, MIRA-Context	recovery accuracy, paired scaffold CI, calls, update cost	VERIFIED
E4b / RQ4	post-call evidence extension	Independent Evidence, Frozen Joint, MIRA planner	accept/call/refuse utility and calibration	OUT OF SCOPE
E4c / RQ4	natural lifecycle scope	provider arrival, retirement, version shift	lifecycle-specific recovery	OUT OF SCOPE
E5 / RQ4	fresh-menu event accounting	all one-call methods	calls, failure/unknown/refusal, state bytes, selection/update p95, recovery	VERIFIED
E6 / RQ4	157 scaffold-disjoint BBB molecules; four fresh checkpoints	Random, Best Fixed, Frozen, Matched SGD, LinUCB, BayesianRouter, MIRA-Context, retrospective best-fixed	selected correctness, paired scaffold CI, calls, update cost	VERIFIED

Table 3: Natural fresh-menu confirmation across 157 scaffold-disjoint molecules. N counts scored decisions and calls counts invocations. Fail/U/R denote execution failures, unknown truth, and refusals. Deltas use the same-receipt Frozen control and scaffold-cluster bootstrap intervals.

Method	N/calls	Success	Paired Δ (95% CI)	Fail/U/R	Update p95
Random (anchor)	4710/4710	83.76%	-6.05 pp [-9.93,-2.26]	0/0/0	2.7 μs
Best Fixed (anchor)	4710/4710	88.15%	-1.66 pp [-3.97,+0.51]	0/0/0	0.5 μs
Frozen (strict)	4710/4710	89.81%	0	0/0/0	0.7 μs
MIRA-Relative	4710/4710	89.81%	0.00 pp [0.00,0.00]	0/0/0	51.5 μs
MIRA-Context	4710/4710	89.81%	0.00 pp [-1.24,+1.40]	0/0/0	1.0 μs
Online SGD (strict)	4710/4710	89.21%	-0.59 pp [-1.80,+0.52]	0/0/0	11.9 μs
Greedy LinUCB	4710/4710	89.72%	-0.08 pp [-0.58,+0.41]	0/0/0	63.5 μs
Bayesian Router	4710/4710	89.53%	-0.28 pp [-1.75,+0.97]	0/0/0	60.5 μs
Retrospective Best Fixed (evaluator-only)	4710/4710	90.45%	+0.64 pp [-2.86,+4.07]	0/0/0	0.5 μs

Table 4: Controlled correctness-drift stress. Changes count post-release action changes relative to Frozen; rescued/harmed counts use the fit-only incumbent. Intervals are scaffold-cluster bootstrap intervals. N counts scored decisions.

Scope / delay	N	Probes	Changes	Rescued / harmed	Paired Δ (95% CI)
Frozen / stress	4710	0	0	—	—
Matched SGD / stress	4710	0	2566	720 / 218	+10.66 pp [+6.05,+15.45]
Greedy LinUCB / stress	4710	0	874	414 / 48	+7.77 pp [+5.08,+10.72]
MIRA-Context / stress	4710	0	1884	780 / 112	+14.18 pp [+9.61,+19.03]

8 Mechanism Interpretation

The two fresh-menu results separate two questions that are often conflated in online routing. The first is whether a provider-specific adapter can change its action after observing a selected result. The second is whether the current provider panel contains a predictable opportunity for that change to improve accuracy. MIRA-Context changes actions in the stationary confirmation, but the 82 rescued and 82 harmed decisions cancel at the aggregate level. This is the expected behavior of a conservative residual: it is allowed to react to evidence without being credited for a gain that the panel does not contain.

Table 5: Reserved evidence-extension table. No one-step accept/call/refuse result is claimed in this paper; the row structure fixes the future utility contract.

Planner	Calls	Refusal rate	System utility	Rescued / harmed
Always accept	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE
Fixed cascade	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE
Independent Evidence	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE
Frozen Joint Evidence	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE
MIRA one-step planner	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE	OUT OF SCOPE

The controlled stress answers the first question directly. At the midpoint, the evaluator flips the returned label only when the fit-best provider is selected. The selector is not told the flip time, the affected provider, or the counterfactual outputs of alternatives. After the delay, released failures accumulate against that provider's residual, and later actions move to providers whose frozen scores are lower but whose observed evidence is now relatively better. Frozen cannot respond because its score surface is immutable. Matched Online-SGD and Greedy LinUCB also respond, which is why they are reported as positive controls rather than straw baselines; MIRA-Context's larger recovery is the result that must survive the common clock and the same feature surface.

The mechanism has a deliberately narrow interpretation. The residual is not a new endpoint predictor and it does not infer a hidden label for an unselected provider. It is a small correction to the fit-time score, centered on the provider's own fit rate. This centering matters in the selected-only setting: an uncentered success count would favor providers that are sampled more often, while a global multiplier would leave an argmax unchanged or change every provider in the same direction. The release ledger makes this distinction observable by recording the provider, due tick, state version, and action sequence for every event.

Three diagnostic comparisons are therefore reported together. Frozen measures the value of the fit-time surface alone. MIRA-Context measures the added value of released selected evidence. MIRA-Relative is a stopped negative variant whose global reliability

factor produces the same action sequence as Frozen on the natural fresh menu. The stress control shows that the positive result is not an artifact of merely having an update field: removing the update path leaves the degraded fit-best provider selected throughout the post-flip segment. The resulting interpretation is an adaptive capability with a measured boundary, rather than a claim that any online update improves a stationary panel.

9 Limitations and Discussion

The benchmark separates public answer semantics from the project routing protocol. This makes the provider-selection question replayable, but it does not make the routing layer a community benchmark by itself. The principal development endpoint is BBB; hERG and AMES remain extensions until their assay semantics, coverage, conflicts, and provider training lineage are closed. The current provider panel also contains related model families, so provider-count improvements cannot be interpreted as independent evidence without family-balanced analysis.

The proposed method is scoped to feedback-adaptive selection. A delayed selected-only replay does not by itself demonstrate natural provider arrival, retirement, version drift, or broad nonstationarity. Those events are controlled stress unless an independent deployment trace supports a stronger claim. The immutable frozen base is a reference and guardrail, not a theorem that no switch can be harmful.

The method may fail when selected feedback is too sparse, when a residual update is too small to change the next choice, or when a provider error transfers to a later decision. These alternatives motivate exposure, score-margin, and rescued/harmed analyses. A failed candidate remains evidence for revising the mechanism; it does not establish the project's target improvement or justify replacing that target with a protocol-only conclusion.

10 Validity and Reproducibility Checks

The confirmation is separated from the development diagnostic at three points. First, provider checkpoints are sealed before the candidate molecules are scored; the four fresh models are bound to their source commit, checkpoint hash, and native prediction code. The 157 confirmation molecules are scaffold-disjoint from the development fit and validation partitions. The truth label is used only by the evaluator after a selected provider returns, never by the selector at decision time.

Second, every stream is recorded as a receipt rather than as an aggregate score. A decision record contains the molecule identifier, menu, feature hash, released-state version, chosen provider, and action score; it contains neither the current truth nor an unselected output. A terminal record binds the chosen provider to its returned output and runtime status. The release queue records the due tick and consumption order. The serialized state, including provider counters and prior count, is hashed after the corresponding selected feedback is released. These fields let the contract/replay verifier reconstruct the provider sequence and detect counter corruption; the verifier reuses runner primitives and is independent of aggregation, not a second implementation.

Third, uncertainty is clustered by molecule scaffold. The natural result pools 5 folds, 2 seeds, and 3 delays over the same 157 confirmation molecules, while the controlled stress uses the same schedule and changes only the selected label of the fit-best provider after tick 78. The reported intervals are bootstrap intervals over Bemis–Murcko scaffold clusters; they are not intervals over individual event rows. We report both rescued and harmed switches because a positive aggregate can hide unsafe substitutions. Contract/replay checks found 270/270 natural streams and 120/120 stress streams replayable with no forbidden decision fields or action mismatches.

This design also fixes what the result cannot establish. The natural fresh menu is a stationary checkpoint panel, so its null online gain is evidence against a generic online improvement claim, not evidence against adaptation under every kind of drift. The stress is a mechanism test with a declared evaluator-side label flip, not a deployment trace. Post-call verification, refusal utility, provider arrival, and the hERG/AMES extensions use separate contracts and are not silently folded into the one-call estimate.

The audit treats the event ledger as part of the method rather than as an implementation detail. A stream is valid only if each decision has one terminal record, one due tick, and one release; if the selected provider does not match the terminal binding, the stream is rejected. The replay then reconstructs the state from the fit prior and the released prefix, and compares both the next provider and the resulting correctness. This catches a particularly easy failure mode in delayed bandits: using a terminal result immediately in the same tick while the aggregate score still appears plausible. The fresh runs use the same canonical molecule order across methods, so paired differences compare action policies on identical inputs and schedules. Seeds affect only policies that sample or initialize an online state; the checkpoint predictions and truth manifest remain fixed. The resulting records are sufficient to audit a single wrong switch, not only to reproduce the final mean.

11 Related Work

Tool-using agents learn to call external APIs and execute actions, but their training objectives generally do not impose a selected-only correctness clock [10, 12]. Our task instead treats the provider call as an online decision whose outcome may be delayed and whose correctness is tied to an independently auditable scientific endpoint.

Contextual bandits provide the natural language for choosing an action from a context before observing its reward. LinUCB [7] and contextual Thompson sampling [1] are canonical controls, while warm-start methods study how prior data can reduce the cost of early exploration [8]. Delayed-feedback analyses show that the release clock changes online learning even when the action space is unchanged [6]. MIRA-Relay uses these ideas as baselines and contracts; its proposed difference is an explicit selected-only release ledger that turns a provider-specific residual into the next action while preserving the scientific endpoint and runtime information boundary.

Recent model-routing work studies static or online selection among language models, including budget-aware adaptive routing [9], graph-based selection [3], dual-contrastive routing [2], and benchmark frameworks for multi-model routing [4]. Those systems provide useful modern comparators, but their usual feedback is a

preference or task score for the selected response and their provider panels are language-model panels. We retain adapted versions only when their input features, prior, feedback, and budget can be audited under the pharmaceutical contract; GraphRouter is audit-only if its graph input semantics cannot be preserved.

Finally, molecular benchmark work emphasizes public task definitions and reproducible datasets [5, 11]. Pharmaceutical providers are correlated predictors with different training lineages, so the present study reports family sensitivity and training-membership limits rather than treating every cached model as an independent truth source.

12 Conclusion

Adapting scientific tool use requires connecting a released outcome to a better subsequent decision. MIRA-Relay makes that connection testable through a frozen provider surface, a selected-only provider residual, and an explicit release clock. On a natural fresh menu it matches the frozen selector; under the declared correctness drift it recovers more accurately than the matched controls. The remaining evidence-extension and lifecycle claims are intentionally outside this paper.

13 Artifact Scope

The released artifact is organized around the routing contract rather than around a single aggregate table. The public truth manifest records canonical molecule identifiers, endpoint labels, split membership, and the hash of the input file. The provider manifest records each fresh checkpoint, its source commit, native feature code, and the validation receipt used to bind the model to a provider identifier. The routing runner then consumes only the molecule features and the current released state; evaluator truth and provider outputs are written to terminal records after the action.

A reproduction has three independent checks. A schema checker verifies that every decision has a legal visible read set and that no decision record contains truth or an unselected output. A transition replayer reconstructs the release queue and next action from the fit prior and the selected event prefix. An aggregation script joins terminal records to the frozen confirmation manifest and computes the paired scaffold-cluster estimates, confidence intervals, rescued/harmed counts, and timing summaries. The checks are run before aggregation, so a malformed stream cannot contribute a partially valid score.

The artifact also preserves negative and stopped variants. MIRA-Relative is retained because its natural fresh result is exactly Frozen; this prevents a global reliability factor from being presented as evidence for dynamic adaptation. The natural MIRA-Context run is retained separately from the controlled correctness-drift run, even though they use the same fresh menu. This separation makes the scope of the positive result inspectable: the adapter responds to a declared released-outcome change, while a stationary panel supplies no measurable aggregate gain.

The current paper intentionally does not claim that the one-call adapter solves post-call verification or refusal. Those interfaces are specified in the contract and remain compatible with the ledger, but they require a second budgeted experiment with their own utility and calibration metrics. The same rule applies to hERG, AMES, provider arrival, retirement, and version changes. Extending the artifact therefore means adding a new frozen receipt and a new gate record, not silently appending rows to the current BBB estimate.

A rerun starts from the sealed input and provider manifest, computes native provider predictions once, and then runs each policy on a copy of the same candidate order. The policy process never receives the truth map; the evaluator process binds the selected prediction to the truth only when it writes the terminal record. Delays are applied by the same queue implementation for every policy, including Frozen and the retrospective best-fixed control, so a method cannot gain by receiving a more favorable clock. The raw event stream is flushed after each stream, which leaves a usable failure boundary if a process stops between streams.

The receipt format separates four kinds of evidence. A decision proves what was visible and which provider was selected. A terminal record proves what the selected provider returned. A feedback record proves when the selected result became legal input to the updater. A stream summary is only a convenience aggregate and is never used as the source of truth for accuracy. The verifier checks identities and cardinalities before reading summaries; it then recomputes selected labels from the sealed checkpoint predictions and the evaluator truth. This is why the same code can be used to audit both the null natural result and the positive stress result without changing their estimands.

The computational budget is also explicit. Every reported stream has one provider call per molecule, 157 decisions, and the same three release delays. The adapter stores two counters per provider, while the frozen surface and checkpoint weights are read-only. Selection cost scales with the number of eligible providers and the fixed feature dimension; release cost is constant in the number of providers. Timing fields are measured around the policy operation and exclude file I/O, provider inference, and evaluator scoring. This decomposition prevents a fast but incomplete update from being compared with a slower method that performs hidden work outside the measured interval.

Finally, the artifact preserves the distinctions needed for later extensions. A runtime error has a status but no chemical-correctness label; an unknown truth ends the scored stream; and a refusal is a policy action with no selected-provider reward. These cases are present in the schema even though the fresh four-provider confirmation contains no such events. Future verification experiments can therefore add accept, call, and refuse actions without redefining the one-call score, while lifecycle experiments can add menu changes without rewriting historical receipts.

References

[1] Shipra Agrawal and Navin Goyal. 2013. Thompson Sampling for Contextual Bandits with Linear Payoffs. In *Proceedings of the 30th International Conference on Machine Learning*. 127–135.

[2] Shuhao Chen, Weisen Jiang, Baijiong Lin, James T. Kwok, and Yu Zhang. 2024. RouterDC: Query-Based Router by Dual Contrastive Learning for Assembling Large Language Models. In *Advances in Neural Information Processing Systems*.

[3] Tao Feng, Yanzhen Shen, and Jiaxuan You. 2025. GraphRouter: A Graph-based Router for LLM Selections. In *International Conference on Learning Representations*.

[4] Qitian Jason Hu, Jacob Bieker, Xiuyu Li, Nan Jiang, Benjamin Keigwin, Gaurav Ranganath, Kurt Keutzer, and Shriyash Kaustubh Upadhyay. 2024. RouterBench: A Benchmark for Multi-LLM Routing System. *arXiv preprint arXiv:2403.12031* (2024).

[5] Kexin Huang, Ting Fu, Lucas M. Glass, Marinka Zitnik, Cao Xiao, and Jimeng Sun. 2021. Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development. In *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*.

[6] Pooria Joulani, Andras Gyorgy, and Csaba Szepesvari. 2013. Online Learning under Delayed Feedback. In *Proceedings of the 30th International Conference on Machine Learning*. 1453–1461.

[7] Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. 2010. A Contextual-Bandit Approach to Personalized News Article Recommendation. In *Proceedings of the 19th International Conference on World Wide Web*. 661–670. doi:10.1145/1772690.1772758

[8] Bastian Oetomo, R. Malinga Perera, Renata Borovica-Gajic, and Benjamin I. P. Rubinstein. 2023. Cutting to the Chase with Warm-Start Contextual Bandits. *Knowledge and Information Systems* 65, 9 (2023), 3533–3565.

[9] Pranoy Panda, Raghav Magazine, Chaitanya Devaguptapu, Sho Takemori, and Vishal Sharma. 2025. Adaptive LLM Routing under Budget Constraints. In *Findings of the Association for Computational Linguistics: EMNLP 2025*. 23934–23949. doi:10.18653/v1/2025.findings-emnlp.1301

[10] Timo Schick, Jane Dwivedi-Yu, Roberto Dessi, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language Models Can Teach Themselves to Use Tools. In *Advances in Neural Information Processing Systems*.

[11] Zhenqin Wu, Bharath Ramsundar, Evan N. Feinberg, Nathan Gomes, Caleb Geniesse, Aneesh S. Pappu, Adele Leswing, and Vijay Pande. 2018. MoleculeNet: A Benchmark for Molecular Machine Learning. *Chemical Science* 9, 2 (2018), 513–530. doi:10.1039/C7SC02664A

[12] Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*.